

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is: The large number of TCP SYN requests coming in from the IP of 203.0.113.0 is leading me to believe the website is witnessing a Denial of service attack called a SYN flood attack. The log items 52-54 indicates that the attacker's SYN request was answered normally by the web server. However, the attacker keeps sending more SYN requests and around log item 125 the web server stopped responding to legitimate employee visitors and the attacker. This event could be a DoS SYN flood attack based off of the number of SYN attempts coming from the specific IP address.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:

1. The SYN packet is the initial request from an employee visitor trying to connect to a web page hosted on the web server.
2. The SYN-ACK packet is the web server's response to the visitor's request agreeing to the connection.
3. The ACK packet is the visitor's machine acknowledging the permission to connect. This is the final step required to make a successful TCP connection.

Explain what happens when a malicious actor sends a large number of SYN packets all at once: When a malicious attacker sends a large number SYN packets all at once it leads to the network being overwhelmed and inaccessible.

Explain what the logs indicate and how that affects the server: The log indicates an unfamiliar IP address is sending an overwhelming amount of SYN requests resulting in the server to stop responding to legitimate employee visitors.